

Django YouTube Clone · Logical Flow

· Project Overview

A YouTube-style video sharing platform with CDN-based video/thumbnaill storage, real-time streaming, and a like/dislike voting system.

· Complete Logical Flow

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·                               Django YouTube Clone                               ·
.....
·
·   ..... VIDEO UPLOAD FLOW .....
·
·   User fills Upload Form
·   (title, description, video file, thumbnail)
·   .
·   .
·   POST · video_upload view
·   .
·   .
·   Validate Form Data
·   .
·   .
·   .....
·   ·   ImageKit CDN Upload
·   ·   ... upload_video(file)
·   ·   ·   ... Returns video_url, file_id
·   ·   ... upload_thumbnail(image)
·   ·   ·   ... Returns thumbnail_url
·   ·   .....
·   .
·   .
·   Create Video Model Instance
·   (title, description, file_id, URLs, views=0)
·   .
·   .
·   Save to Database
·   .
·   ..... VIDEO VIEWING FLOW .....
·
·   User navigates to video list
·   .
·   .
·   video_list view ... Display feed
·   .
·   .
·   Click video ... video_detail view
·   .
·   ·   ... Increment views += 1
·
```

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.      ... Get streaming URL from ImageKit .
.      ... Get thumbnail with watermark .
.      ... Check user's existing vote .
.      . .
.      .
. Render Video Player .
.      .
.      ..... VOTING FLOW ..... .
.      .
. User clicks Like/Dislike .
.      .
.      .
. POST video_vote view .
.      .
.      .
.      .....
.      . Check Existing Vote .
.      . .
.      . .....
.      . .
.      . .
.      . No vote exists      Vote exists .
.      . .
.      . .
.      . Create new          Same value? .
.      . VideoLike          ... Yes . Remove .
.      . record              ... No . Flip .
.      . .
.      . .....
.      . .
.      . Adjust like/dislike counters .
.      . (atomic update) .
.      . .....
.      .
.      .
. Return JSON { likes, dislikes, user_vote } .
.      .
.      .....

```